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ORIGINAL RESEARCH

Estimating causal effects on quality of life under treatment discontinuation: the ALTA-1L trial

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Abstract

Background and Objectives: Time to worsening in health-related quality of life (HRQoL) is increasingly used in oncology trials. Treatment discontinuation poses a challenge: once discontinued, patients cease HRQoL assessments, precluding outcome observation. Standard survival analyses censor at discontinuation, assuming noninformative censoring—an assumption often violated when discontinuation relates to disease progression or toxicity. Bridging the ICH E9(R1) estimand framework with causal inference methods clarifies how to define and estimate treatment effects in such settings. We reanalyze time to worsening in global health status from the ALTA-1L trial (brigatinib vs crizotinib in ALK + non−small-cell lung cancer), integrating both frameworks.

Methods: Following Young et al's (2020) causal framework for competing events, we defined two estimands structured according to ICH E9(R1): (1) a controlled direct effect under a hypothetical strategy, envisioning a scenario where treatment discontinuation would not occur, estimated using inverse probability of censoring weighted Kaplan-Meier to adjust for informative censoring; and (2) a total effect under a while-on-treatment strategy, with discontinuation as a competing event, estimated using the Aalen-Johansen estimator. Risk ratios (RRs) were estimated at 36 months with bootstrap CIs.

Results: The original ALTA-1L analysis reported hazard ratio = 0.69 (95% CI: 0.49, 0.98), censoring at discontinuation and assuming noninformative censoring. Deriving the RR at 36 months from Kaplan-Meier curves yields 0.75 (95% CI: 0.59, 0.97). After adjusting for informative censoring, the controlled direct effect was RR = 0.89 (95% CI: 0.65, 1.26). The total effect was RR = 1.03 (95% CI: 0.76, 1.40), reflecting the competing risk structure: earlier discontinuation in the crizotinib arm (discontinuation RR = 0.54; 95% CI: 0.38, 0.72) reduced observed worsening events. These different estimates illustrate how different estimands address distinct clinical questions.

Conclusion: This study bridges the ICH E9(R1) estimand framework with causal inference methods for time-to-event HRQoL analysis when discontinuation precludes observation. By quantifying bias from standard approaches, we provide methodological clarity for applied researchers. To facilitate practical implementation, we translate these insights into a decision flowchart for estimand specification and method selection when intercurrent events (ICEs) act as competing events. Future trials should prespecify ICE-handling strategies and consider data collection beyond ICEs to support treatment policy estimation.

Keywords: Cancer; Survival analysis; Health-related quality of life; Competing events; Causal effects

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Trial registration: Clinicaltrials.gov Identifier: NCT02737501 on April 14, 2016.

Ethics statement: The ALTA-1L trial was conducted in accordance with the principles of the Declaration of Helsinki and the International Council for Harmonisation guidelines for good clinical practice. Written informed consent was obtained from all study participants for data sharing. For our research work, we submitted a waiver approval to the Ethics Committee at the University of Munich. The committee thoroughly evaluated the provided documentation and information to ensure adherence to the ethical principles outlined in the Declaration of Helsinki, Section 15 of the professional code, and faculty law. The Ethics Committee approved the study (Project No.: 22-0934) and raised no objections to its conduct.

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What is new?

Key findings

- Using data from the ALTA-1L trial, we demonstrate how conventional survival-based analyses can overstate health-related quality of life (HRQoL) benefit when treatment discontinuation differs between arms and leads to cessation of further HRQoL assessment.
- Different strategies for handling treatment discontinuation define distinct estimands (clinical questions), resulting in different interpretations of HRQoL benefit.

What this adds to what is known?

- We integrate the ICH E9(R1) estimand framework with causal-inference principles to clarify HRQoL interpretation under treatment discontinuation; and propose a qualitative decision flowchart to align estimands with analytical methods.

What is the implication and what should change now?

- Improvements in time-to-event end points (eg, progression-free survival) do not necessarily translate into improved HRQoL; rigorous HRQoL evaluation requires prespecified estimands and transparent handling of treatment discontinuation, supported by appropriate trial design.
- Existing trials should be reexamined to assess the robustness of HRQoL conclusions when treatment discontinuation and other postrandomization events are explicitly accounted for.

1. Introduction

1.1. Health-related quality of life in oncology trials

Patient-reported outcomes (PROs) provide essential insights into patients' symptoms, functioning, and overall health status [1−3]. In oncology trials, PROs are commonly assessed using health-related quality of life (HRQoL) measures [4], typically as secondary or exploratory end points [2,5] to complement clinical outcomes [6]. In advanced nonsmall cell lung cancer (NSCLC) trials, where survival gains from targeted therapies and immunotherapies are accompanied by prolonged treatment exposure, HRQoL evaluation becomes increasingly important [4], informing regulatory decisions [1,2,5,7,8], health technology assessments [9], and clinical practice. Despite their relevance, HRQoL end points are often analyzed without clearly defined research questions [9], risking misinterpretation and limiting their use in clinical practice [1,2,10−12]. This lack of clarity becomes particularly acute for time-to-event outcomes such as time to worsening in HRQoL in the presence of intercurrent events (ICEs)—postrandomization events that affect either outcome interpretation or existence [13,14]. For such outcomes, ICEs manifest in two distinct ways [15]: semicompeting events that alter the hazard of the outcome without precluding its observation or competing events that preclude observation altogether. Treatment discontinuation—common in oncology due to disease progression, toxicity, or protocol requirements [9]—is a frequent ICE that acts as a competing event when patients stop attending clinic visits, thereby precluding further HRQoL assessment [5,16,17].

1.2. Causal inference, the ICH E9(R1) estimand framework, and competing events

Randomized controlled trials (RCTs) provide the gold standard for causal inference, recovering treatment effects via all causal mechanisms [18]. When ICEs such as treatment discontinuation act as competing events, the interpretation of the resulting treatment effects depends on the precise causal question being investigated [19]. While statistical methods for addressing treatment effects in the presence of competing events are well established [20], the focus has largely been on estimation rather than on precisely defining the target of estimation—the estimand [21]. Addressing this gap, Young et al [22] introduced a counterfactual framework for causal inference in competing event settings that distinguishes between two causal estimands: a total effect, which captures all causal pathways between treatment and the outcome, including those operating through the competing event, and a controlled direct effect, representing the treatment effect under a hypothetical intervention that eliminates the competing event.

In parallel, the ICH E9(R1) addendum introduced the estimand framework to address misalignment between clinical trial objectives and reported treatment effects [14,23], providing a structured approach for formulating precise research questions through explicit estimand definition [1,3,5,24,25], which comprises five attributes: population, treatment, outcome, ICE−handling strategy, and summary measure. The addendum also outlines five strategies for handling ICEs: treatment policy, composite, hypothetical, while-on-treatment, and principal stratum, each reflecting a different clinical question.

Within the ICH E9(R1) framework, ICE−handling strategies define how postrandomization events are addressed, whereas causal estimands formalize the corresponding treatment effects implied by each strategy. Despite their shared objective of clarifying treatment effects in the presence of ICEs, these two frameworks have largely developed in parallel, with limited to no reference to one another [13].

Young et al formalize causal estimands for competing events without explicit linkage to ICH E9(R1), while the estimand framework articulates clinically meaningful strategies for handling ICEs without formally grounding them in counterfactual causal settings. This lack of integration represents a missed opportunity to align regulatory ICE-handling strategies and estimand definitions with causal interpretations, particularly for time-to-event data, where the choice of ICE strategy has direct implications for identifiability and causal interpretation [15].

1.3. Aims and motivating example

This paper bridges the ICH E9(R1) estimand framework with causal inference methods to address analytical challenges posed by treatment discontinuation in time to worsening analyses of HRQoL. Using data from the ALTA-1L trial—a randomized, open-label, phase 3 study comparing brigatinib and crizotinib in ALK-positive NSCLC [26,27]—we demonstrate how these frameworks can be integrated when discontinuation acts as a competing event. The original analysis reported a hazard ratio (HR) of 0.69 (95% CI: 0.49, 0.98) for time to worsening in global health status (GHS), censoring patients at treatment discontinuation—a standard approach mandated by the 2007 Food and Drug Administration Cancer Endpoint Guidance [28] and still widely implemented in oncology trials. This approach assumes noninformative censoring, which may be questionable when treatment discontinuation is associated with disease progression or treatment-related toxicity.

We reanalyze this end point by defining two causal estimands—following Young et al [22]—and aligning them with ICH E9(R1) ICEs−handling strategies: a controlled direct effect estimated using inverse probability of censoring weighting (IPCW) under a hypothetical strategy that eliminates treatment discontinuation, and a total effect estimated using competing risks methods under a while-on-treatment strategy, which defines the outcome up to treatment discontinuation. We additionally present a decision flowchart illustrating how explicit handling of ICEs informs method selection for time-to-event HRQoL analysis.

2. Materials and methods

2.1. Directed acyclic graph (DAG) and target causal estimands

Figure 1 presents a simplified DAG illustrating the causal structure underlying our analyses. The DAG represents the PRO population—patients with baseline and at least 1 postbaseline GHS assessment—and reflects key features of the ALTA-1L trial design and data structure. Five variable types are depicted: treatment assignment at randomization (Z), baseline and time-varying covariates (TVCs; L_k), treatment discontinuation (D_k), GHS worsening (W_k), and unmeasured factors (U). Treatment may affect GHS worsening through two pathways: directly ($Z \rightarrow W_k$), representing the treatment effect independent of discontinuation, and indirectly ($Z \rightarrow D_k \rightarrow W_k$), capturing effects mediated through differential discontinuation between arms. The bold arrow from $D_k \rightarrow W_k$ denotes the competing risk structure—once discontinued, patients cannot experience on-treatment GHS worsening at subsequent time points. The dotted arrow ($U \rightarrow D_k$) represents possible unmeasured confounding of discontinuation, with implications for estimand identification discussed below. Loss to follow-up is excluded from the DAG, as it occurred predominantly after treatment discontinuation, and is assumed to occur at random.

We define a primary and a supplementary estimand, each structured according to the ICH E9(R1) framework (Table 1) and motivated by the competing risks causal inference framework [22]:

- Primary estimand (controlled direct effect): Following a hypothetical strategy [16], this estimand represents the treatment effect under hypothetical

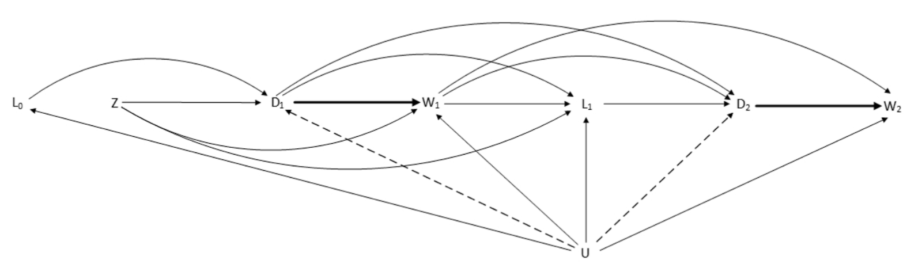

Figure 1. Directed acyclic graph (DAG) representing the assumed causal relationships in the ALTA-1L HRQoL data, guiding our analyses of time to worsening in GHS in the presence of treatment discontinuation. The DAG is simplified to baseline ($k = 0$) and two follow-up intervals ($k = 1,2$). Key variables include treatment assignment at randomization (Z), baseline and time-varying covariates (L_k), treatment discontinuation (D_k), and worsening in global health status (W_k). Treatment assignment Z is unconfounded by design. This DAG highlights that treatment Z affects worsening in GHS W_k through both a direct path (from Z to W_k) and an indirect one via treatment discontinuation (Z to W_k through D_k). The bold arrow from $D_k \rightarrow W_k$ indicates the competing risk structure, where discontinuation precludes subsequent on-treatment GHS worsening. Unmeasured factors (U) affect GHS worsening W_k through either measured covariates L_k, treatment discontinuation D_k, or directly. The dotted arrow ($U \rightarrow D_k$) represents a possible direct effect of unmeasured factors U on discontinuation D_k. Loss to follow-up (LTFU) is excluded from the diagram, as censoring due to LTFU occurred predominantly after treatment discontinuation; we assume LTFU is missing at random. GHS, global health status.

Table 1. Estimand attributes

Target estimand	Estimand attributes	Estimand
Controlled direct effect on time to worsening in GHS *(Primary estimand)*	Treatment	Brigatinib (orally at a dose of 90 mg QD for 7 days, then 180 mg QD) vs crizotinib (250 mg taken orally BID).
	Population	PRO population[a]
	End point	Time to worsening in GHS[b]
	ICE-handling strategy	Hypothetical strategy[c]
	Population-level summary measure	Ratio of cumulative incidence at 36 months
Total effect on time to worsening in GHS *(Supplementary estimand)*	Treatment	Same as above
	Population	PRO population[a]
	End point	Time to worsening in GHS[b]
	ICE-handling strategy	While-on-treatment strategy[d]
	Population-level summary measure	Ratio of cumulative incidence at 36 months

BID, twice daily; GHS, global health status; ICE, intercurrent event; PRO, patient-reported outcome; QD, once daily.
[a] The ALTA-1L population included patients who met the eligibility criteria, with analyses restricted to those with baseline and at least 1 post-baseline PRO data point.
[b] Worsening in GHS defined as a $\geq$ 10 point decrease from baseline in the GHS scale, measured using EORTC-QLQ-C30.
[c] Isolates the direct treatment pathway ($Z \rightarrow W_k$ in Fig 1) under a hypothetical strategy, envisioning elimination of treatment discontinuation. Estimated using inverse probability of censoring weighted Kaplan—Meier (IPCW—KM).
[d] Captures all causal pathways: direct effect and indirect effect through discontinuation ($Z \rightarrow W_k$ and $Z \rightarrow D_k \rightarrow W_k$ in Fig 1). Estimated using Aalen—Johansen estimator.

elimination of the competing event [22], isolating the direct pathway $Z \rightarrow W_k$. It compares cumulative incidences of GHS worsening at 36 months if all patients had remained on assigned treatment: "always treated with brigatinib" ($Z = 1, \overline{D} = \overline{0}$) vs "always treated with crizotinib" ($Z = 0, \overline{D} = \overline{0}$). Identification requires consistency, exchangeability, and positivity (Table 2), where exchangeability requires that all common causes of treatment discontinuation and GHS worsening are adequately captured by measured covariates (no direct $U \rightarrow D_k$ path). This assumption is supported by the comprehensive covariate data in ALTA-1L. While the original ALTA-1L analysis implemented a hypothetical strategy by censoring at treatment discontinuation, our IPCW approach [29,30] targets a risk-based controlled direct effect and explicitly adjusts for measured confounders of discontinuation, enabling assessment of bias arising from unadjusted informative censoring.

- Supplementary estimand (total effect): Following a while-on-treatment strategy [16], this estimand captures the overall effect of treatment assignment when discontinuation precludes outcome observation, incorporating both the direct pathway ($Z \rightarrow W_k$) and indirect pathway through discontinuation ($Z \rightarrow D_k \rightarrow W_k$) [22]. It compares cumulative incidences of GHS worsening before discontinuation at 36 months between "assigned to brigatinib" (Z=1) and "assigned to crizotinib" (Z=0), with discontinuation treated as a competing event. This effect remains identified even with unmeasured confounding of discontinuation (dotted $U \rightarrow D_k$ arrow), as it does

Table 2. Identifiability conditions for estimating total and controlled direct treatment effects

Identifiability conditions	Total treatment effect	Controlled direct treatment effect
Positivity	Hold by design of an RCT	We assume a nonzero probability of remaining on treatment at each time point within strata of treatment assignment and covariate history; violations lead to undefined or unstable IPCW weights
Exchangeability	Hold by design of an RCT	The effect of all unmeasured confounders of both the outcome (worsening in GHS) and the competing event (treatment discontinuation) are adequately captured through the measured covariates.
Consistency	Hold by design of an RCT	The hypothetical intervention eliminating treatment discontinuation must be clearly defined.

GHS, global health status; RCT, randomized controlled trial.

Table 3. Types of analyses in the presence of competing events and their respective quantities

Approach	Quantities	Method
Marginal analysis		
Causal	Marginal cumulative incidence[a]	Kaplan-Meier estimator
Noncausal	Marginal hazard[a]	Cox PH model
Competing risk analysis		
Causal	Marginal cumulative incidence under a hypothetical strategy[b] (controlled direct effect)	IPCW-weighted KM estimator
	Cause-specific cumulative incidence under a while-on-treatment strategy[c] (total effect)	Aalen-Johansen estimator
Noncausal	Cause-specific hazard	Cause-specific Cox PH model
	Subdistributional hazard	Fine and Gray model

Cox PH, Cox proportional hazard; IPCW, inverse probability of censoring weighting; KM, Kaplan-Meier.
[a] Requires that the competing event and outcome are independent.
[b] Requires conditional independence of discontinuation and outcome given measured confounders; weights adjust for informative censoring.
[c] Requires that censoring due to loss to follow-up and outcome are independent.

not require modeling D_k. As patients who discontinue earlier may appear to have a lower risk of worsening, we also assessed discontinuation rates to contextualize this effect.

2.2. Approaches to survival analysis with competing events

Table 3 outlines common approaches for estimating treatment effects in the presence of competing events. These approaches can be broadly distinguished by (1) whether they target marginal or competing-risk quantities and (2) whether the resulting effect measures have a causal interpretation under explicit assumptions. Common noncausal measures include marginal HRs from Cox proportional hazard models, cause-specific hazard ratios from cause-specific Cox models, and subdistribution hazard ratios from Fine−Gray models. These hazard-based measures have limited causal interpretability due to proportional hazards assumptions and built-in selection bias [31]. This paper focuses on estimating causal effects using risk ratios (RRs) at a fixed time point corresponding to the estimands defined in Section 2.3. A glossary distinguishing these effect measures and related terminology is provided in Table S1.

2.3. Estimating controlled direct effect and total effect

To assess reproducibility of the published ALTA-1L HRQoL analysis, we first reproduced the Cox proportional hazards model for time to worsening in GHS reported in the original trial publication, censoring follow-up at treatment discontinuation and thereby relying on a noninformative censoring assumption. This analysis implements a hypothetical strategy through censoring but targets a hazard-based effect measure. To enable direct comparison with our causal estimands defined in Subsection 2.1, which are expressed as RRs at 36 months, we converted the reproduced HR to 36-month RRs using (1) the Breslow estimator of the baseline cumulative hazard and (2) Kaplan−Meier estimates censoring at treatment discontinuation. These reproducibility-based RRs are reported alongside the causal estimates in Table 4.

To estimate the controlled direct effect under a hypothetical strategy, treatment discontinuation was treated as a censoring event. We applied IPCW to adjust for informative censoring and estimate the marginal cumulative incidence of GHS worsening that would have been observed had all patients remained on their assigned treatment. Each individual at each time point was assigned a weight inversely proportional to their estimated probability of remaining on treatment, conditional on baseline and TVCs. Referring to the DAG (Fig 1), IPCW corresponds to removing arrows from treatment assignment (Z) and TVCs (L_k) to treatment discontinuation events (D_k, D_{k+1}), ensuring exchangeability conditional on observed covariates. Weights were estimated separately by treatment arm following the method of Robins and Finkelstein [29]. The numerator was derived from a pooled logistic regression model based on time; the denominator included time, baseline covariates, and TVCs (full covariate list in Table S2. Covariates were prespecified based on prior literature on treatment switching predictors [27] and selected using either a P value threshold ($\leq$0.20) or OR $\leq$0.75/$\geq$1.33) or Akaike Information Criterion. We then estimated the marginal cumulative incidence of GHS worsening using the complement of a weighted Kaplan−Meier estimator, adapted from Saunero et al (2023) [21]. RRs at 36 months were derived with percentile bootstrap CIs (1000 samples). Identifiability of the controlled direct effect requires additional assumptions beyond randomization—positivity, conditional exchangeability, and consistency—which are summarized in Table 2. Full regression results,

Table 4. Controlled direct effect and total effect of brigatinib vs crizotinib on time to worsening in GHS; ALTA-1L trial

Estimand and implementation	Risk (95% CI)[a] In crizotinib	Risk (95% CI)[a] In brigatinib	RR (95% CI)[a,b]
Original ALTA-1L (reproducibility)[c]			
Cox PH-based[d]	0.65 (0.55, 0.75)	0.52 (0.43, 0.62)	0.80 (0.64, 0.99)
KM-based[e]	0.68 (0.56, 0.79)	0.51 (0.41, 0.60)	0.75 (0.59, 0.97)
Controlled direct treatment effect, primary estimand			
IPCW-weighted KM[f]	0.58 (0.45, 0.72)	0.52 (0.41, 0.62)	0.89 (0.65, 1.26)
Total treatment effect, supplementary estimand			
Aalen-Johansen[g]	0.40 (0.32, 0.50)	0.42 (0.33, 0.50)	1.03 (0.76, 1.40)

Cox PH, Cox proportional hazards; GHS, Global Health Status; IPCW, inverse probability of censoring weighting; KM, Kaplan-Meier; RR, risk ratio.

[a] 95% CIs estimated using nonparametric bootstrap (1000 samples).

[b] RR: ratio of cumulative incidences at 36 months.

[c] Original ALTA-1L analysis reported HR = 0.69 (95% CI: 0.49, 0.98), reproduced as HR = 0.70 (95% CI: 0.49, 0.99), censoring at discontinuation and assuming noninformative censoring. We derived RRs at 36 months using Cox PH and KM approaches to enable comparison with our causal estimates.

[d] RR derived from reproduced Cox PH model (HR = 0.70 (95% CI: 0.49, 0.99). Cumulative risks: 1-exp(-$H_0(t)$) (control) and 1-exp(-$H_0(t) \times$HR) (experimental), where $H_0(t)$ is Breslow baseline cumulative hazard.

[e] RR derived directly from KM curves censoring at discontinuation.

[f] IPCW-adjusted controlled direct effect comparing ''always treated with brigatinib'' vs ''always treated with crizotinib''. Adjusts for informative censoring at discontinuation.

[g] Total effect comparing ''assigned to brigatinib'' vs ''assigned to crizotinib.'' RR of time to treatment discontinuation is 0.54 (95% CI: 0.38, 0.72).

weight diagnostics, and sensitivity analyses are provided in Tables S3−S4, Fig S1.

To estimate the total effect of treatment on time to worsening in GHS, we treated treatment discontinuation as a competing event and estimated cause-specific cumulative incidences using the Aalen−Johansen estimator. This analysis corresponds to a while-on-treatment strategy [16], capturing both direct effects of treatment on GHS worsening and indirect effects mediated through differential treatment discontinuation. RRs at 36 months were calculated from the estimated cumulative incidences. The identifiability conditions—consistency, positivity, and exchangeability—are expected to hold by design in randomized controlled trials (Table 2).

3. Results

3.1. Reproducing ALTA-1L trial results

The ALTA-1L trial reported an HR of 0.69 (95% CI: 0.49, 0.98) from a Cox PH model for time to worsening in GHS, censoring at treatment discontinuation and assuming noninformative censoring. We successfully reproduced this result (HR = 0.70; 95% CI: 0.49−0.99). To enable direct comparison with our causal estimates expressed as RRs (Sections 3.2 and 3.3), we converted the reproduced HR to RRs and examined their stability over time using Kaplan−Meier curves. RRs remained stable between 12 and 36 months (range: 0.72−0.75) but diverged at 48 months (RR = 0.64) due to data sparsity in the crizotinib arm Table S5. We therefore selected 36 months as the primary time point for all subsequent analyses, balancing precision with sufficient follow-up for differential discontinuation patterns to manifest. At 36 months, RRs were estimated as 0.80 (95% CI: 0.64, 0.99) using the Breslow baseline hazard estimator and 0.75 (95% CI: 0.59, 0.97) directly from Kaplan−Meier estimates, both assuming noninformative censoring (Table 4).

3.2. Controlled direct effect on time to worsening in GHS

The controlled direct effect, comparing ''always treated with brigatinib'' to ''always treated with crizotinib'', was estimated using an IPCW-weighted Kaplan−Meier estimator under a hypothetical strategy envisioning continued treatment. At 36 months, the RR was 0.89 (95% CI: 0.65, 1.26) compared to the unadjusted KM-based estimate of 0.75 (95% CI: 0.59, 0.97)—a shift toward the null that quantifies the bias from the noninformative censoring assumption in the original analysis. Results from this analysis were consistent between stabilized and unstabilized weights. Sensitivity analyses across four model specifications also demonstrated robustness (RR range: 0.86−0.89). Main results are presented in Table 4 and Figure 2A; weight distributions, sensitivity analyses, and model specifications are detailed in Tables S3−S4, Fig S1.

3.3. Total effect on time to worsening in GHS

The total effect, comparing ''assigned to brigatinib'' to ''assigned to crizotinib'' on time to worsening in GHS,

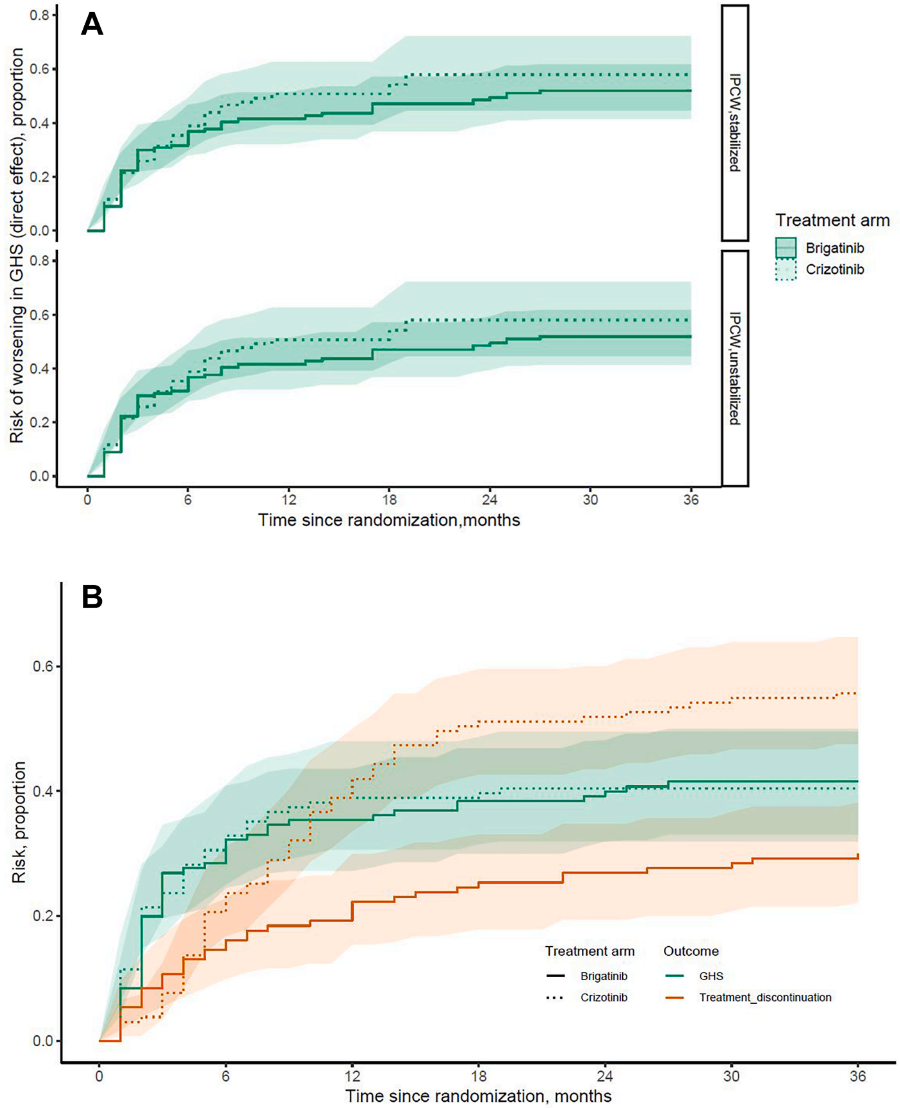

Figure 2. Survival curves displaying total and controlled direct effects.

was estimated using the Aalen–Johansen estimator under a while-on-treatment strategy. At 36 months, the RR was 1.03 (95% CI: 0.76, 1.40). Time to treatment discontinuation differed substantially between arms (RR = 0.54; 95% CI: 0.38, 0.72), with earlier discontinuation in the crizotinib group. This differential discontinuation contextualizes the total effect through discontinuation: patients on crizotinib who discontinued earlier are no longer observed, reducing the chance of detecting worsening. Results are presented in Table 4 and Figure 2B.

4. Discussion

This work was motivated by the limited interpretability of the published ALTA-1L HRQoL findings in the presence of treatment discontinuation. The original analysis censored patients at discontinuation and reported a HR for time to worsening in GHS, implicitly assuming noninformative censoring and disregarding the relationship between stopping treatment and HRQoL. In oncology trials where discontinuation is common and closely linked to toxicity, disease progression, or trial design features, this approach obscures the clinical question being addressed. Guided by the ICH E9(R1) estimand framework and the competing events causal framework of Young et al., we reanalyzed these data using two clearly defined estimands that reflect distinct clinical questions.

The controlled direct effect, corresponding to a hypothetical strategy in which treatment discontinuation is eliminated, estimated an RR of 0.89 (95% CI: 0.65, 1.26) using IPCW. Compared with the unadjusted Kaplan–Meier estimate (RR = 0.75), the attenuation toward the null illustrates the impact of informative censoring when discontinuation is not independent of HRQoL worsening. The wide CI reflects both the modest PRO sample size

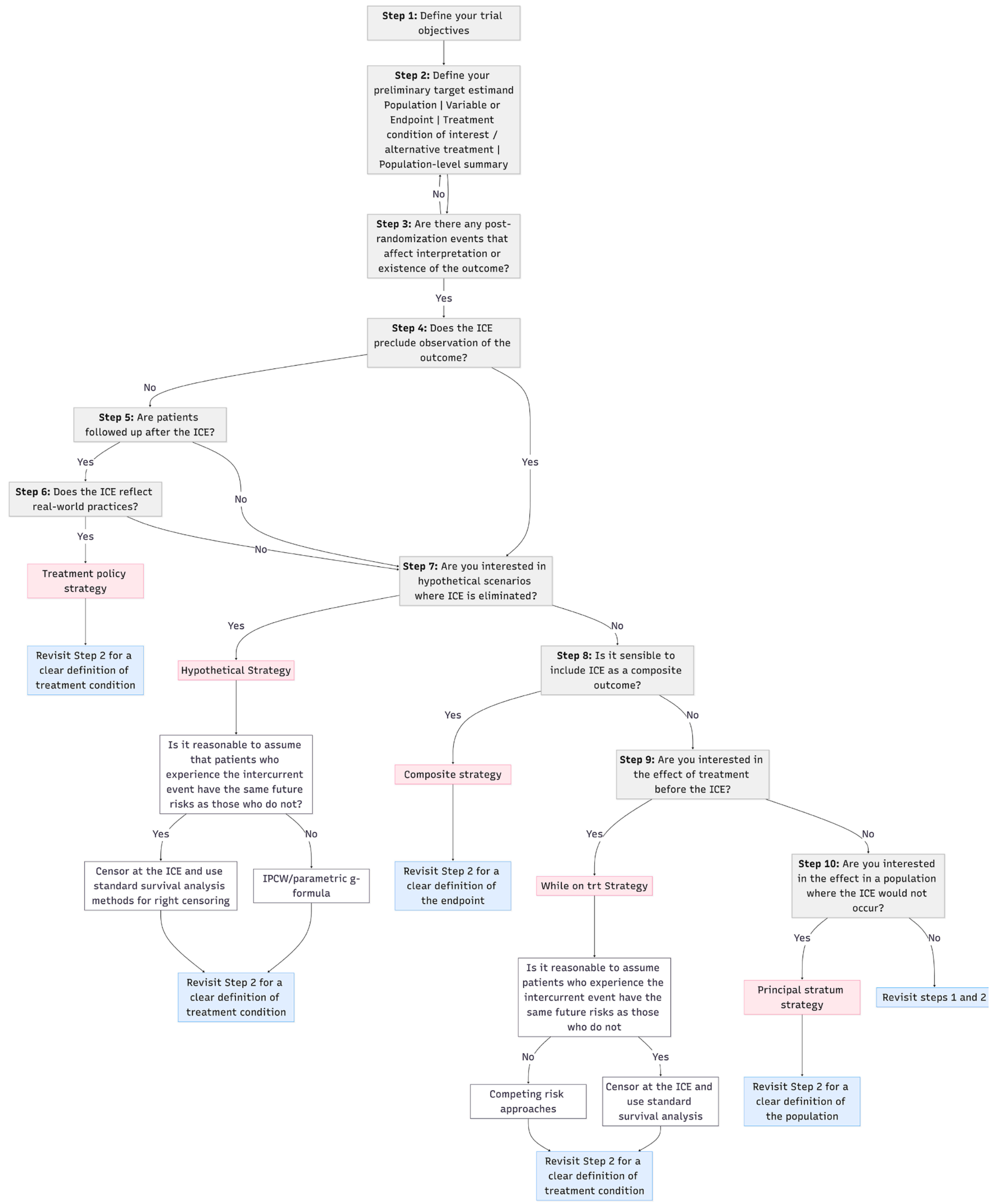

Figure 3. Decision flowchart: An iterative framework for operationalizing ICH E9(R1) estimand attributes in time-to-event analysis. A decision guide with application from the ALTA-1L trial.

and the efficiency loss inherent to weighting approaches. Identification of this estimand requires strong assumptions, including a well-defined hypothetical intervention and adequate control of confounding. While hypothetical elimination of discontinuation has been criticized for limited clinical relevance, in ALTA-1L many discontinuations arose from potentially preventable causes, such as adverse events that may be mitigated with improved toxicity management—through dose modifications, supportive care, or patient education—or trial-specific crossover not present in routine care. In this context, reducing the influence of trial-specific and potentially preventable sources of discontinuation allows the controlled direct effect to approximate HRQoL under sustained treatment exposure as observed in routine clinical practice. Separable effects [32] represent a promising methodological alternative for future work.

In contrast, the total effect under a while-on-treatment strategy yielded an RR of 1.03 (95% CI: 0.76, 1.40), reflecting the competing-risk structure of the data. Patients receiving crizotinib discontinued earlier (RR for treatment discontinuation = 0.54), thereby contributing less observed time at risk for HRQoL worsening. This near-null estimate does not indicate equivalence between treatments, but rather demonstrates how differential discontinuation can mask treatment differences when outcome observation ceases after treatment stopping. In ALTA-1L, death before discontinuation or worsening was rare and evenly distributed (6 out of 262 patients); however, in settings with higher mortality or longer follow-up, death should be modeled explicitly as a separate competing event.

These HRQoL findings should be interpreted in the broader context of end point discordance in ALTA-1L. While brigatinib demonstrated a clear progression-free survival benefit [27], overall survival advantages have been reported as plausible but not fully convincing [33]. In this reanalysis, neither the controlled direct effect nor the total effect provides compelling evidence of an HRQoL advantage for brigatinib. Importantly, the pronounced difference in treatment discontinuation suggests relevant differences in tolerability requiring further exploration, as discontinuation in ALTA-1L was influenced not only by toxicity but also by trial-specific features such as crossover [33−35] in the crizotinib arm. Our findings highlight that favorable results on earlier or surrogate end points do not automatically translate into improvements in patient-reported quality of life, underscoring the importance of estimand clarity and cautious interpretation when informing clinical decision-making.

Methodologically, this work bridges a gap between the ICH E9(R1) estimand framework and causal inference approaches for competing events. By explicitly aligning the hypothetical strategy with the controlled direct effect and the while-on-treatment strategy with the total effect, we demonstrate how regulatory estimand concepts can be operationalized using causal methods. Whether the while-on-treatment strategy fully captures the notion of a total effect, or whether additional estimand categories are needed, warrants further investigation.

To support applied researchers, we propose a qualitative decision flowchart (Fig 3) to guide estimand specification and method selection when ICEs act as competing events. The flowchart emphasizes that strategy choice fundamentally reframes the clinical question and should be considered during trial design. While quantitative benchmarks for decision nodes remain an area for future methodological research, this flowchart can aid prospective planning by anticipating data collection and analysis challenges.

Several limitations should be noted. Our reanalysis is based on a single trial and 1 HRQoL domain, limiting generalizability. Future work should validate this methodological framework across different tumor types with varying natural histories and discontinuation patterns, as well as alternative HRQoL domains. Intermittent missing data, though minimal ($<2\%$), were handled as provided by the sponsor to preserve comparability with the original analysis. Monthly assessments introduce interval censoring, though this affects both arms equally and is accommodated by discrete-time analysis. Treatment policy estimands, though representing pragmatically what happens in a population when one treatment is replaced by another one, could not be evaluated because HRQoL assessment ceased at discontinuation, highlighting a structural limitation when ICEs preclude outcome observation. We encourage greater adoption of the estimand framework in HRQoL analyses and emphasize the importance of prospectively planning data collection to support valid estimation—including continued outcome assessment after key ICEs when treatment policy effects are of interest.

5. Conclusion

This study illustrates how causal inference methods can be integrated with the ICH E9(R1) estimand framework to address treatment discontinuation in time-to-event HRQoL analyses. Clearly defined causal estimands—a controlled direct effect under a hypothetical strategy and a total effect under a while-on-treatment strategy—correspond to distinct clinical questions with different assumptions. Our findings underscore the importance of explicit estimand specification when ICEs affect outcome observation. Future work may extend these principles to other outcome types facing similar challenges.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the author(s) used ChatGPT (OpenAI) in order to improve language and readability. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

CRediT authorship contribution statement

Amani Al Tawil: Writing − original draft, Visualization, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Michael Lauseker:** Writing − review & editing, Methodology. **Ulrich Mansmann:** Writing − review & editing, Supervision, Methodology, Conceptualization.

Declaration of competing interest

There are no competing interests for any author.

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Supplementary data

Supplementary data related to this article can be found at https://doi.org/10.1016/j.jclinepi.2026.112189.

Data availability

The dataset used in this study was obtained from the Takeda-sponsored ALTA-1L trial via the Vivli data-sharing platform (https://vivli.org/) through a controlled access mechanism. Access is granted upon approval by an independent review panel to ensure participant privacy is protected. Researchers interested in accessing the data should submit a request directly to Vivli. The analysis code can be provided by the corresponding author upon request.

References

[1] Fiero MH, Pe M, Weinstock C, King-Kallimanis BL, Komo S, Klepin HD, et al. Demystifying the estimand framework: a case study using patient-reported outcomes in oncology. Lancet Oncol 2020;21(10):e488−94.

[2] Fiero MH, Roydhouse JK, Bhatnagar V, Chen TY, King-Kallimanis BL, Tang S, et al. Time to deterioration of symptoms or function using patient-reported outcomes in cancer trials. Lancet Oncol 2022;23(5):e229−34.

[3] Bell ML, Floden L, Rabe BA, Hudgens S, Dhillon HM, Bray VJ, et al. Analytical approaches and estimands to take account of missing patient-reported data in longitudinal studies. Patient Relat Outcome Meas 2019;10:129−40.

[4] Waisberg F, Lopez C, Enrico D, Rodriguez A, Hirsch I, Burton J, et al. Assessing the methodological quality of quality-of-life analyses in first-line non-small cell lung cancer trials: a systematic review. Crit Rev Oncol Hematol 2022;176:103747.

[5] Lawrance R, Skaltsa K, Regnault A, Floden L. Reflections on estimands for patient-reported outcomes in cancer clinical trials. J Biopharm Stat 2023;35:1−11.

[6] Jensen RE, Moinpour C, Fairclough DL. Assessing health-related quality of life in cancer trials. Clin Invest 2012;2:563−77.

[7] Gnanasakthy A, Russo J, Gnanasakthy K, Harris N, Castro C. A review of patient-reported outcome assessments in registration trials of FDA-approved new oncology drugs (2014-2018). Contemp Clin Trials 2022;120:106860.

[8] Teixeira MM, Borges FC, Ferreira PS, Rocha J, Sepodes B, Torre C. A review of patient-reported outcomes used for regulatory approval of oncology medicinal products in the European Union between 2017 and 2020. Front Med (Lausanne) 2022;9:968272.

[9] King-Kallimanis BL, Calvert M, Cella D, Cocks K, Coens C, Fairclough D, et al. Perspectives on patient-reported outcome data after treatment discontinuation in cancer clinical trials. Value Health 2023;26(10):1543−8.

[10] Van Der Weijst L, Lievens Y, Schrauwen W, Surmont V. Health-related quality of life in advanced non-small cell lung cancer: a methodological appraisal based on a systematic literature review. Front Oncol 2019;9:715.

[11] Kyte D, Retzer A, Ahmed K, Keeley T, Armes J, Brown JM, et al. Systematic evaluation of patient-reported outcome protocol content and reporting in cancer trials. J Natl Cancer Inst 2019;111(11):1170−8.

[12] Hamel JF, Saulnier P, Pe M, Zikos E, Musoro J, Coens C, et al. A systematic review of the quality of statistical methods employed for analysing quality of life data in cancer randomised controlled trials. Eur J Cancer 2017;83:166−76.

[13] Keene ON, Lynggaard H, Englert S, Lanius V, Wright D. Why estimands are needed to define treatment effects in clinical trials. BMC Med 2023;21(1):276.

[14] European Medicines Agency. ICH E9 (R1) addendum on estimands and sensitivity analysis in clinical trials to the guideline on statistical principles for clinical trials. Amsterdam, The Netherlands: EMA/CHMP/ICH/436221/2017, Committee for Medicinal Products for Human Use; 2020.

[15] Deng Y, Han S, Zhou XH. Inference for Cumulative Incidences and Treatment Effects in Randomized Controlled Trials With Time-to-Event Outcomes Under ICH E9 (R1) [Journal Article]. Stat Med 2025;44(10-12):e70091.

[16] Siegel JM, Weber HJ, Englert S, Liu F, Casey M. Time-to-event estimands and loss to follow-up in oncology in light of the estimands guidance. Pharm Stat 2024;23(5):709−27.

[17] Unkel S, Amiri M, Benda N, Beyersmann J, Knoerzer D, Kupas K, et al. On estimands and the analysis of adverse events in the presence of varying follow-up times within the benefit assessment of therapies. Pharm Stat 2019;18(2):166−83.

[18] Robins JM, Greenland S. Identifiability and exchangeability for direct and indirect effects. Epidemiology 1992;3(2):143−55.

[19] Gabriel EE, Ocampo A, Sjölander A. Elucidating some common biases in randomized controlled trials using directed acyclic graphs. Eur J Epidemiol 2025.

[20] Geskus RB. Competing risks: concepts, methods, and software. Annu Rev Stat Its Appl 2024;11:227−54.

[21] Rojas-Saunero LP, Young JG, Didelez V, Ikram MA, Swanson SA. Considering questions before methods in dementia research with competing events and causal goals. Am J Epidemiol 2023;192(8):1415−23.

[22] Young JG, Stensrud MJ, Tchetgen Tchetgen EJ, Hernán MA. A causal framework for classical statistical estimands in failure-time settings with competing events. Stat Med 2020;39(8):1199−236.

[23] Rufibach K. Treatment effect quantification for time-to-event endpoints-Estimands, analysis strategies, and beyond. Pharm Stat 2019;18(2):145−65.

[24] Lawrance R, Degtyarev E, Griffiths P, Trask P, Lau H, D'Alessio D, et al. What is an estimand & how does it relate to quantifying the effect of treatment on patient-reported quality of life outcomes in clinical trials? J Patient Rep Outcomes 2020;4(1):68.

[25] Kahan BC, Hindley J, Edwards M, Cro S, Morris TP. The estimands framework: a primer on the ICH E9(R1) addendum. BMJ 2024;384:e076316.

[26] Camidge DR, Kim HR, Ahn MJ, Yang JC, Han JY, Lee JS, et al. Brigatinib versus crizotinib in ALK-positive non-small-cell lung cancer. N Engl J Med 2018;379(21):2027−39.

[27] Camidge DR, Kim HR, Ahn MJ, Yang JCH, Han JY, Hochmair MJ, et al. Brigatinib versus crizotinib in ALK inhibitor-naive advanced ALK-positive NSCLC: final results of phase 3 ALTA-1L trial. J Thorac Oncol 2021;16(12):2091−108.

[28] United States Food and Drug Administration. Clinical Trial Endpoints for the Approval of Cancer Drugs and Biologics: Guidance for Industry 2007: Rockville, Maryland.

[29] Robins JM, Finkelstein DM. Correcting for noncompliance and dependent censoring in an AIDS clinical trial with inverse probability of censoring weighted (IPCW) log-rank tests. Biometrics 2000;56(3):779−88.

[30] Cole SR, Hernán MA. Constructing inverse probability weights for marginal structural models. Am J Epidemiol 2008;168(1):656−64.

[31] Hernán MA. The hazards of hazard ratios. Epidemiology 2010;21(1):13−5.

[32] Stensrud MJ, Young JG, Didelez V, Robins JM, Hern'an MA. Separable effects for causal inference in the presence of competing events. J Am Stat Assoc 2019;117:175−83.

[33] Al Tawil A, McGrath S, Ristl R, Mansmann U. Addressing treatment switching in the ALTA-1L trial with g-methods: exploring the impact of model specification. BMC Med Res Methodol 2024;24(1):314.

[34] Sullivan TR, Latimer NR, Gray J, Sorich MJ, Salter AB, Karnon J. Adjusting for treatment switching in oncology trials: a systematic review and recommendations for reporting. Value Health 2020;23(3):388−96.

[35] Latimer NR. Treatment switching in oncology trials and the acceptability of adjustment methods. Expert Rev Pharmacoecon Outcomes Res 2015;15(4):561−4.